

Appendix B. Fuzzy Operators and Properties

This appendix presents the most common operators used in fuzzy logic literature and some noteworthy properties [28, 37, 64, 69].

Appendix B.1. Negation

Definition 7. A negation is a function $N : [0, 1] \rightarrow [0, 1]$ that at least satisfies:

- N1. Boundary conditions: $N(0) = 1$ and $N(1) = 0$,
- N2. Monotonically decreasing: $\forall (x, y) \in [0, 1]^2, x \leq y \rightarrow N(x) \geq N(y)$.

Moreover, a negation is said to be *strict* if N is continuous and strictly decreasing. A negation is said to be *strong* if $\forall x \in [0, 1], N(N(x)) = x$.

We commonly use the standard strict and strong negation $N_S(a) = 1 - a$.

Appendix B.2. Conjunction

Definition 8. A conjunction is a function $C : [0, 1]^2 \rightarrow [0, 1]$ that at least satisfies:

- C1. boundary conditions: $C(0, 0) = C(0, 1) = C(1, 0) = 0$ and $C(1, 1) = 1$,
- C2. monotonically increasing: $\forall (x, y, z) \in [0, 1]^3$, if $x \leq y$, then $C(x, z) \leq C(y, z)$ and $C(z, x) \leq C(z, y)$.

In fuzzy logic, t-norms are widely used to model conjunction operators.

Definition 9. A t-norm (triangular norm) is a function $t : [0, 1]^2 \rightarrow [0, 1]$ that at least satisfies:

- T1. boundary conditions: $T(x, 1) = x$,
- T2. monotonically increasing,
- T3. commutative,
- T4. associative.

Example 4. Three commonly used t-norms are:

$$\begin{aligned} T_M(x, y) &= \min(x, y) && \text{(minimum)} \\ T_P(x, y) &= x \cdot y && \text{(product)} \\ T_L(x, y) &= \max(x + y - 1, 0) && \text{(\u0141ukasiewicz)} \end{aligned}$$

Name	$a \wedge b$	$a \vee b$	$a \rightarrow_R c$	$a \rightarrow_S c$
Goedel	$\min(a, b)$	$\max(a, b)$	$\begin{cases} 1, & \text{if } a \leq c \\ c, & \text{otherwise} \end{cases}$	$\max(1 - a, c)$
Goguen/Product	$a \cdot b$	$a + b - a \cdot b$	$\begin{cases} 1, & \text{if } a \leq c \\ \frac{c}{a}, & \text{otherwise} \end{cases}$	$1 - a + a \cdot c$
\u0141ukasiewicz	$\max(a + b - 1, 0)$	$\min(a + b, 1)$	$\min(1 - a + c, 1)$	$\min(1 - a + c, 1)$

Table B.4: Common Symmetric Configurations

Appendix B.3. Disjunction

Definition 10. A disjunction is a function $D : [0, 1]^2 \rightarrow [0, 1]$ that at least satisfies:

- D1. boundary conditions: $D(0, 0) = 0$ and $D(0, 1) = D(1, 0) = D(1, 1) = 1$,
- D2. monotonically increasing: $\forall (x, y, z) \in [0, 1]^3$, if $x \leq y$, then $D(x, z) \leq D(y, z)$ and $D(z, x) \leq D(z, y)$.

Disjunctions in fuzzy logic are often modeled with t-conorms.

Definition 11. A t-conorm (triangular conorm) is a function $S : [0, 1]^2 \rightarrow [0, 1]$ that at least satisfies:

- S1. boundary conditions: $S(x, 0) = x$,
- S2. monotonically increasing,
- S3. commutative,
- S4. associative.

Example 5. Three commonly used t-conorms are:

$$\begin{aligned} S_M(x, y) &= \max(x, y) && \text{(maximum)} \\ S_P(x, y) &= x + y - x \cdot y && \text{(probabilistic sum)} \\ S_L(x, y) &= \min(x + y, 1) && \text{(Łukasiewicz)} \end{aligned}$$

Note that the only distributive pair of t-norm and t-conorm is T_M and S_M – that is, distributivity of the t-norm over the t-conorm, and inversely.

Definition 12. The N -dual t-conorm S of a t-norm T w.r.t. a strict fuzzy negation N is defined as:

$$\forall (x, y) \in [0, 1]^2, S(x, y) = N(T(N(x), N(y))). \quad (\text{B.1})$$

If N is a strong negation, we also get:

$$\forall (x, y) \in [0, 1]^2, T(x, y) = N(S(N(x), N(y))). \quad (\text{B.2})$$

Appendix B.4. Implication

Definition 13. An implication is a function $I : [0, 1]^2 \rightarrow [0, 1]$ that at least satisfies:

- I1. boundary Conditions: $I(0, 0) = I(0, 1) = I(1, 1) = 1$ and $I(1, 0) = 0$

Definition 14. There are two main classes of implications generated from the fuzzy logic operators for negation, conjunction and disjunction.

S-Implications *Strong implications* are defined using $x \rightarrow y = \neg x \vee y$ (*material implication*).

R-Implications *Residuated implications* are defined using $x \rightarrow y = \sup\{z \in [0, 1] \mid x \wedge z \leq y\}$. One way of understanding this approach is a generalization of *modus ponens*: the consequent is at least as true as the (fuzzy) conjunction of the antecedent and the implication.

Example 6. Popular fuzzy implications and their classes are presented in Table B.5.

Name	$I(x, y) =$	S-Implication	R-Implication
Kleene-Dienes I_{KD}	$\max(1 - x, y)$	$S = S_M$ $N = N_S$	-
Goedel I_G	$\begin{cases} 1, & x \leq y \\ y, & \text{otherwise} \end{cases}$	-	$T = T_M$
Reichenbach I_R	$1 - x + xy$	$S = S_P$ $N = N_S$	-
Goguen I_P	$\begin{cases} 1, & x \leq y \\ y/x, & \text{otherwise} \end{cases}$	-	$T = T_P$
Łukasiewicz I_{Luk}	$\min(1 - x + y, 1)$	$S = S_L$ $N = N_S$	$T = T_L$

Table B.5: Popular fuzzy implications and their classes. Strong implications (S-Implications) are defined using a fuzzy negation and fuzzy disjunction. Residuated implications (R-Implications) are defined using a fuzzy conjunction.

Appendix B.5. Aggregation

Definition 15. An aggregation operator is a function $A : \bigcup_{n \in \mathbb{N}} [0, 1]^n \rightarrow [0, 1]$ that at least satisfies:

- A1. $A(x_1, \dots, x_n) \leq A(y_1, \dots, y_n)$ whenever $x_i \leq y_i$ for all $i \in \{1, \dots, n\}$,
- A2. $A(x) = x$ for all $x \in [0, 1]$,
- A3. $A(0, \dots, 0) = 0$ and $A(1, \dots, 1) = 1$.

Example 7. Candidates for universal quantification $\forall$ can be obtained using t-norms with $A_T(x_i) = x_i$ and $A_T(x_1, \dots, x_n) = T(x_1, A_T(x_2, \dots, x_n))$:

$$A_{T_M}(x_1, \dots, x_n) = \min(x_1, \dots, x_n) \quad (\text{minimum})$$

$$A_{T_P}(x_1, \dots, x_n) = \prod_{i=1}^n x_i \quad (\text{product})$$

$$A_{T_L}(x_1, \dots, x_n) = \max\left(\sum_{i=1}^n x_i - n + 1, 0\right) \quad (\text{Łukasiewicz})$$

Similarly, candidates for existential quantification $\exists$ can be obtained using s-norms with $A_S(x_i) = x_i$ and $A_S(x_1, \dots, x_n) = S(x_1, A_S(x_2, \dots, x_n))$:

$$A_{S_M}(x_1, \dots, x_n) = \max(x_1, \dots, x_n) \quad (\text{maximum})$$

$$A_{S_P}(x_1, \dots, x_n) = 1 - \prod_{i=1}^n (1 - x_i) \quad (\text{probabilistic sum})$$

$$A_{S_L}(x_1, \dots, x_n) = \min\left(\sum_{i=1}^n x_i, 1\right) \quad (\text{Łukasiewicz})$$